

Demand Forecasting



Approach



Factors Affecting Sales Forecasting



Demand Metrics



Seasonality



Trend



Randomness

Events



Competitor Product Launch



Price Change



*Forecasted Sales**

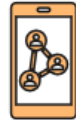
Marketing Activities



Promotions Discounts



Digital Campaigns



Media Activities

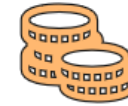
External Factors



Holidays



Weekend

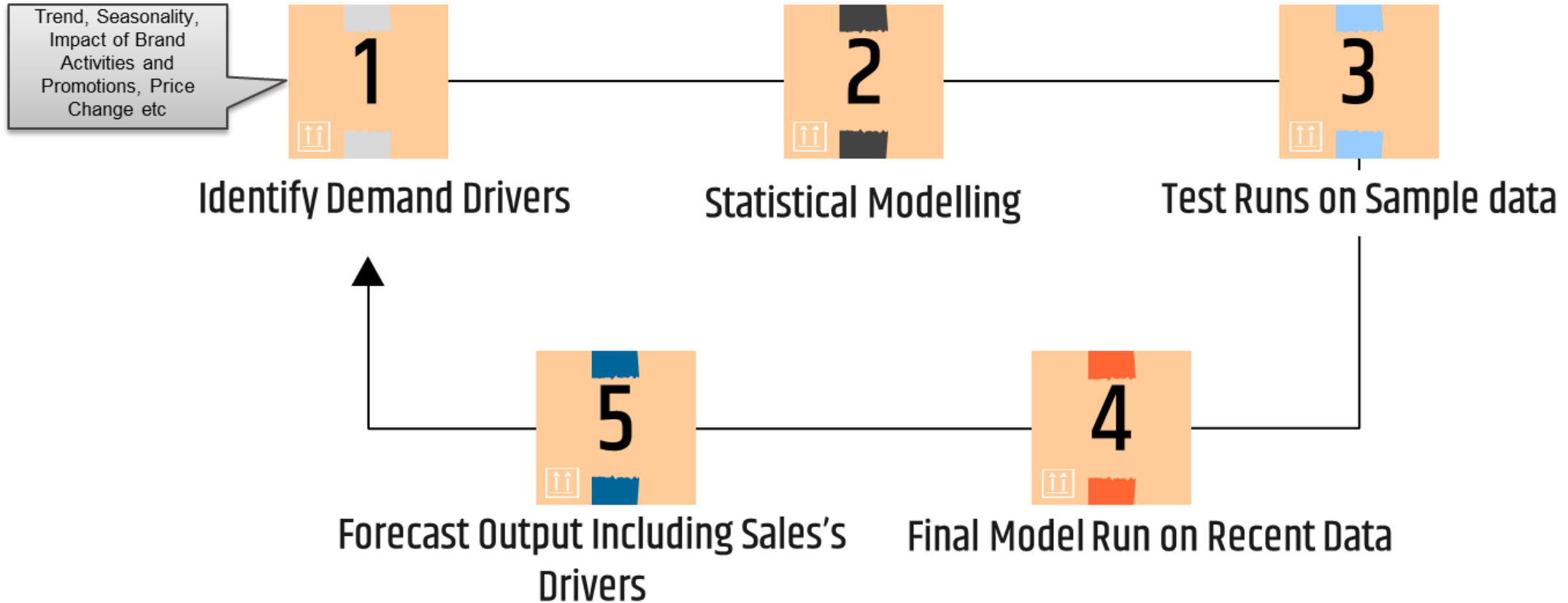


Macro-Economic Factors



Weather Patterns

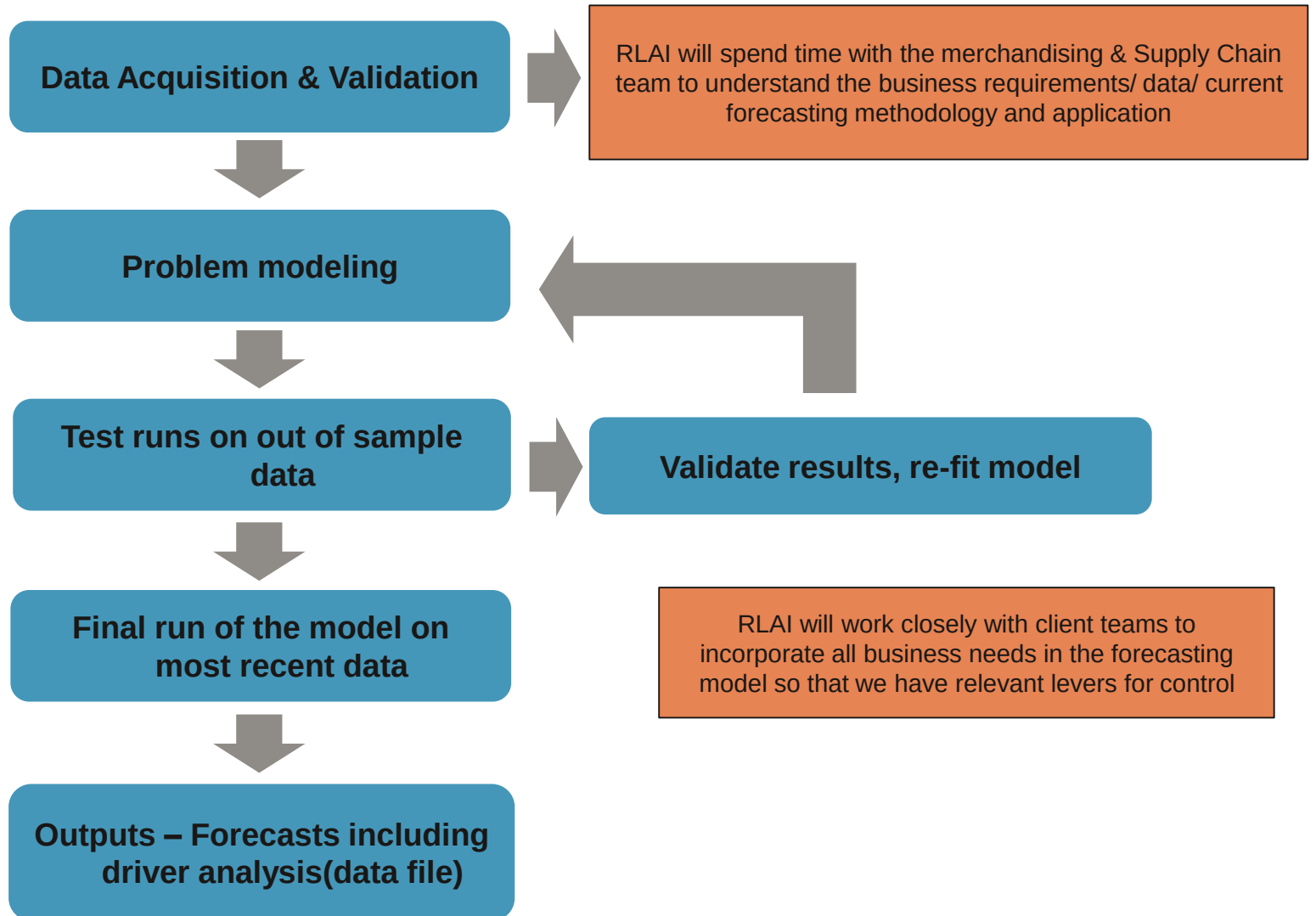
Sales and Strategy Plan



METHODOLOGY DEMAND FORECASTING

The Methodology chosen will need to capture the following elements:

- Trend
- Seasonality
- Impact of brand activities/ marketing
- Impact of promotions (no data)
- Macroeconomic factors
- Prices (no data)



DIFFERENT METHODOLOGIES CONSIDERED



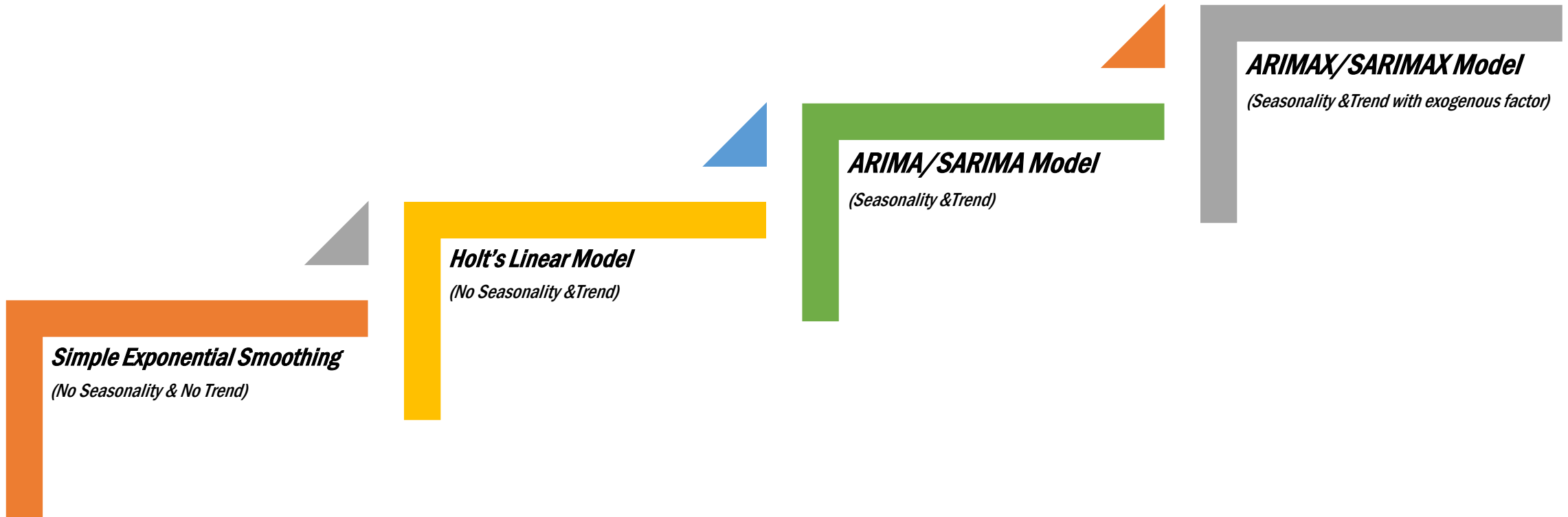
Additional approaches

- Due to pandemic scenario, forecasted values based on ARIMA method had started showing increasing bias for few of the product categories, as these are some uncertain trends observed in previous 2 years.
- We applied alternate methods to forecast the demand for these product categories, to provide an insight to supply chain teams and stores have an optimal stock of the required category before the festive season between Oct – Dec 2021.
- The heuristic methods are used, and final results are based on the combination of 2 or more methods which will factor in multiple trends and scenarios
 - 1, 2, 3-year MA
 - % Jump from M_{t-1} to M_t
 - % Jump from M_{t-12} to M_t
 - M_t as multiple of M_{t-11} sales
 - Jump in Average Daily Sales from M_{t-1} post offer to M non offer LY
 - Jump in ADS from M_{t-12} to M_t Non offer LY
 - % Jump in ADB from M_{t-12} to M_t Non offer LY
 - % Jump in ADB from M_{t-12} to M_t Non offer LY
 - % Jump in ADB Nov non offer wrt 2019
- Final Projected- Avg of method X,Y,&Z for various Product categories

Note:

1. With the given data, we can model the data using ARIMAX methodology

Methodology



State Space Models

- A linear-Gaussian state space model for an m -dimensional time series y_t consists of a measurement equation relating the observed data to an p -dimensional state vector θ_t , and a Markovian transition equation that describes the evolution of the state vector over time

The measurement equation has the form

$$\underset{(m \times 1)}{y_t} = \underset{(m \times p)}{F_t} \underset{(p \times 1)}{\theta_t} + \underset{(m \times 1)}{v_t},$$

$$v_t \sim \text{iid } N(0, V_t)$$

The transition equation for the state vector θ_t is the first order Markov process

$$\underset{(p \times 1)}{\theta_t} = \underset{(p \times p)}{G_t} \underset{(p \times 1)}{\theta_{t-1}} + \underset{(p \times 1)}{w_t}$$

$$w_t \sim \text{iid } N(0, W_t) \quad E[v_t w_s] = 0 \text{ for all } s, t = 1, \dots, T$$

ARIMAX Method

- An ARIMAX model can be viewed as a multiple regression model with one or more autoregressive (AR) terms and/or one or more moving average (MA) terms
- Autoregressive terms for a dependent variable are merely lagged values of that dependent variable that have a statistically significant relationship with its most recent value
- Moving average terms are nothing more than residuals (i.e., lagged errors) resulting from previously made estimates
- Example: A nameless time-series dependent variable might be well estimated by a properly weighted combination of the following four right-hand-side (RHS) variables

$$Y_t = b_1 X_t + f_1 Y_{t-1} + f_2 Y_{t-2} + q_1 e_{t-4}$$

- x_t = The value of the independent variable “ x ” at time “ t ”
- Y_{t-1} = The immediately preceding value of the dependent variable at time “ $t-1$ ”
- Y_{t-2} = the immediately preceding value of the dependent variable at time “ $t-2$ ”
- e_{t-4} = the estimation error produced by the model at time “ $t-4$ ”

b_1, f_1, f_2 and q_1 are estimated coefficients

DRIVERS OF SALES

Analysis Process

STAGE I

Data Preparation

- ▶ Identify appropriate granularity and time frame for analysis
- ▶ Aggregate sales data at appropriate level of granularity
- ▶ Create the dependent and the independent variables at the required level of granularity

STAGE II

Exploratory Data Analysis & Model development

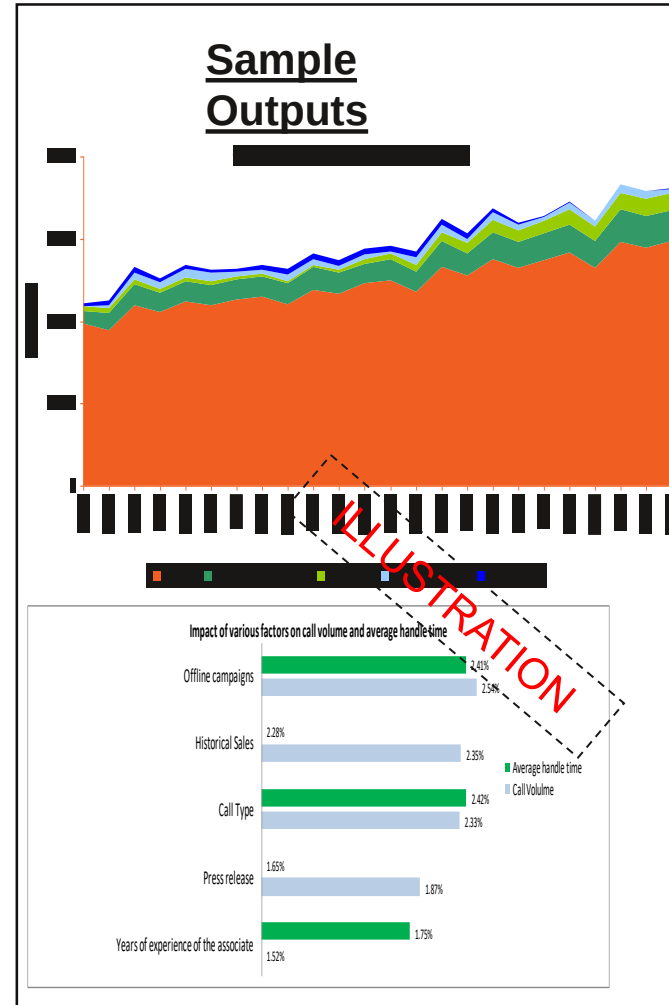
- EDA
- ▶ Perform missing value and outlier treatment
 - ▶ Univariate and Bivariate analysis
 - ▶ Understand correlations and test key hypotheses
- Model framework
- ▶ Multivariate regression to understand the drivers of sales volume

STAGE III

Business Interpretation and Findings Consolidation

- ▶ Understand variables that are part of the model and those that are not
- ▶ Determine contributions of the significant variables
- ▶ Generate actionable insights
- ▶ Tailor recommendations based on insights

Sample Outputs



Results

(Training Period – (Jan'2018-Sep'2023))

Item	Error %
HN-8202M	8%
HS1-3001H	19%
SW-0686Z	15%
TD-5009UA	12%
RV-5006RD	7%
HL-5205M	4%
OF3012SP	18%

Forecast

	Forecast	Actual	Forecast	Forecast
SKU	Jan'2024	Jan'2024	Feb'2024	Mar'2024
HL-5205M	19,623	18,000	22,044	27,471
HN-8202M	2,55,194	2,62,480	2,57,350	1,87,819
HS1-3001H	2,22,056	2,62,600	1,71,637	1,68,095
OF3012SP	30,759	31,616	31,598	30,053
RV-5006RD	31,077	29,400	53,454	56,304
SW-0686Z	1,52,635	1,63,640	1,69,305	1,43,509
TD-5009UA	95,212	80,680	1,38,345	1,26,139

* Forecast is done on Order Qty